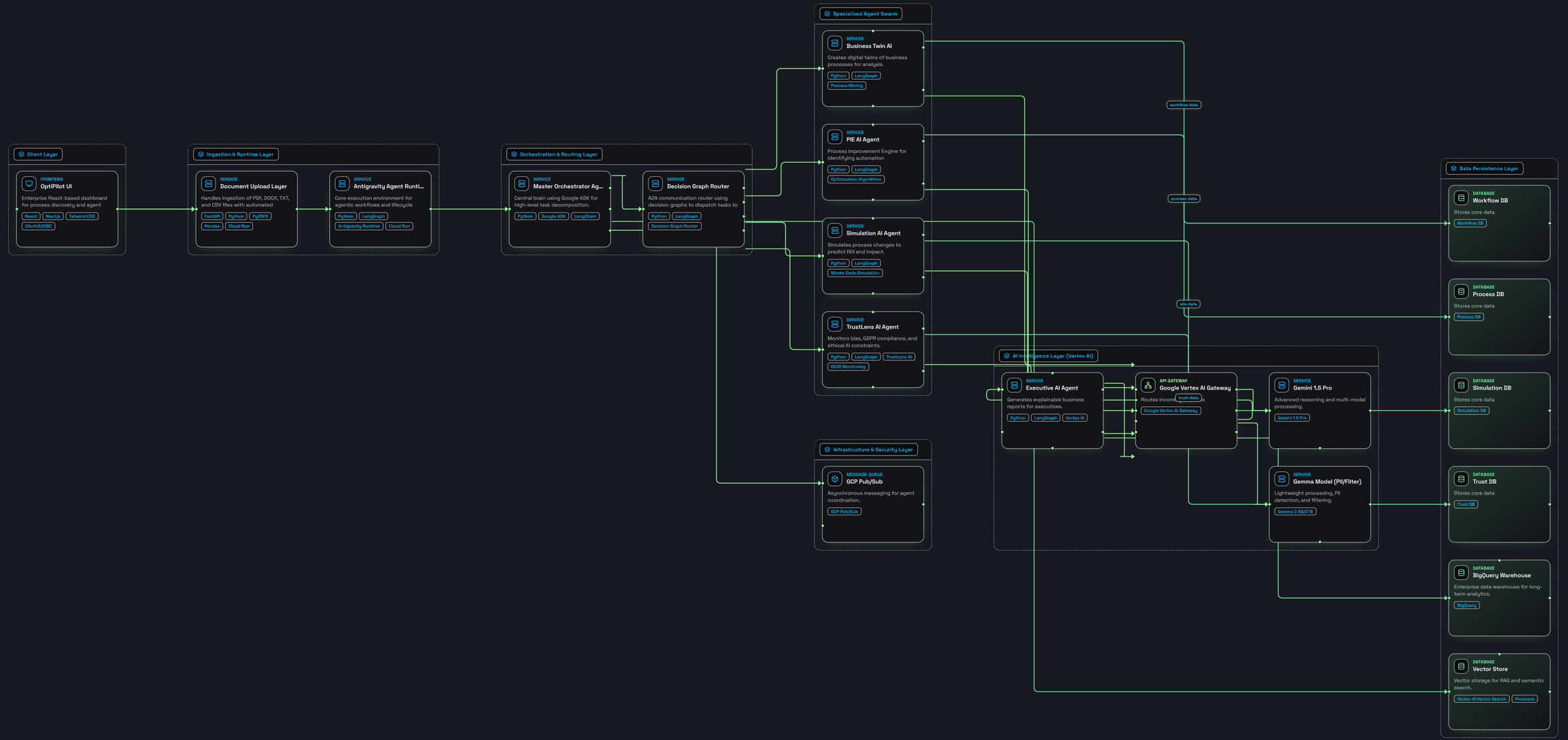
Client Layer
FRONTEND
OptiPilot UI
Enterprise React-based dashboard for process discovery and agent
React
Next.js
Tailwind CSS
OAuth2/OIDC
Ingestion & Runtime Layer
SERVICE
Document Upload Layer
Handles ingestion of PDF, DOCX, TXT, and CSV files with automated
FastAPI
Python
PyPDF2
Pandas
Cloud Run
SERVICE
Antigravity Agent Runti...
Core execution environment for agentic workflows and lifecycle
Python
LangGraph
Antigravity Runtime
Cloud Run
Orchestration & Routing Layer
SERVICE
Master Orchestrator Ag...
Central brain using Google ADK for high-level task decomposition.
Python
Google ADK
LangChain
SERVICE
Decision Graph Router
A2A communication router using decision graphs to dispatch tasks to
Python
LangGraph
Decision Graph Router
Specialized Agent Swarm
SERVICE
Business Twin AI
Creates digital twins of business processes for analysis.
Python
LangGraph
Process Mining
SERVICE
PIE AI Agent
Process Improvement Engine for identifying automation
Python
LangGraph
Optimization Algorithms
SERVICE
Simulation AI Agent
Simulates process changes to predict ROI and impact.
Python
LangGraph
Monte Carlo Simulation
SERVICE
TrustLens AI Agent
Monitors bias, GDPR compliance, and ethical AI constraints.
Python
LangGraph
TrustLens AI
EEOC Monitoring
Infrastructure & Security Layer
MESSAGE-QUEUE
GCP Pub/Sub
Asynchronous messaging for agent coordination.
GCP Pub/Sub
AI Intelligence Layer (Vertex AI)
SERVICE
Executive AI Agent
Generates explainable business reports for executives.
Python
LangGraph
Vertex AI
API GATEWAY
Google Vertex AI Gateway
Routes incom
trust-data
Google Vertex AI Gateway
SERVICE
Gemini 1.5 Pro
Advanced reasoning and multi-modal processing.
Gemini 1.5 Pro
SERVICE
Gemma Model (PII/Filter)
Lightweight processing, PII detection, and filtering.
Gemma 2 9B/27B
workflow-data
process-data
sim-data
Data Persistence Layer
DATABASE
Workflow DB
Stores core data
Workflow DB
DATABASE
Process DB
Stores core data
Process DB
DATABASE
Simulation DB
Stores core data
Simulation DB
DATABASE
Trust DB
Stores core data
Trust DB
DATABASE
BigQuery Warehouse
Enterprise data warehouse for long-term analytics.
BigQuery
DATABASE
Vector Store
Vector storage for RAG and semantic search.
Vertex AI Vector Search
Pinecone